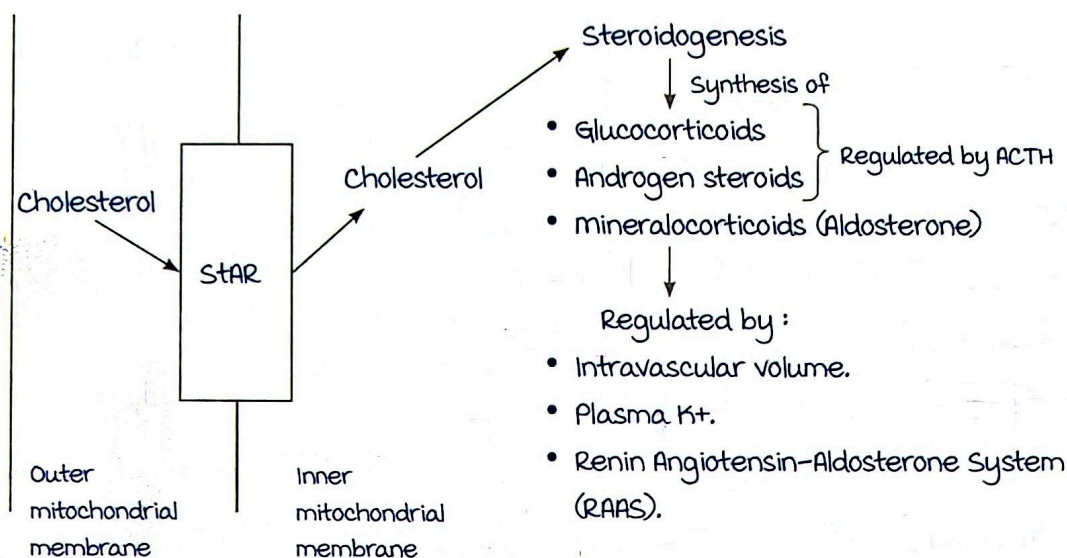


ADRENAL DISORDERS IN CHILDREN

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PHYSIOLOGY:



STAR : Steroidogenic acute regulatory protein.

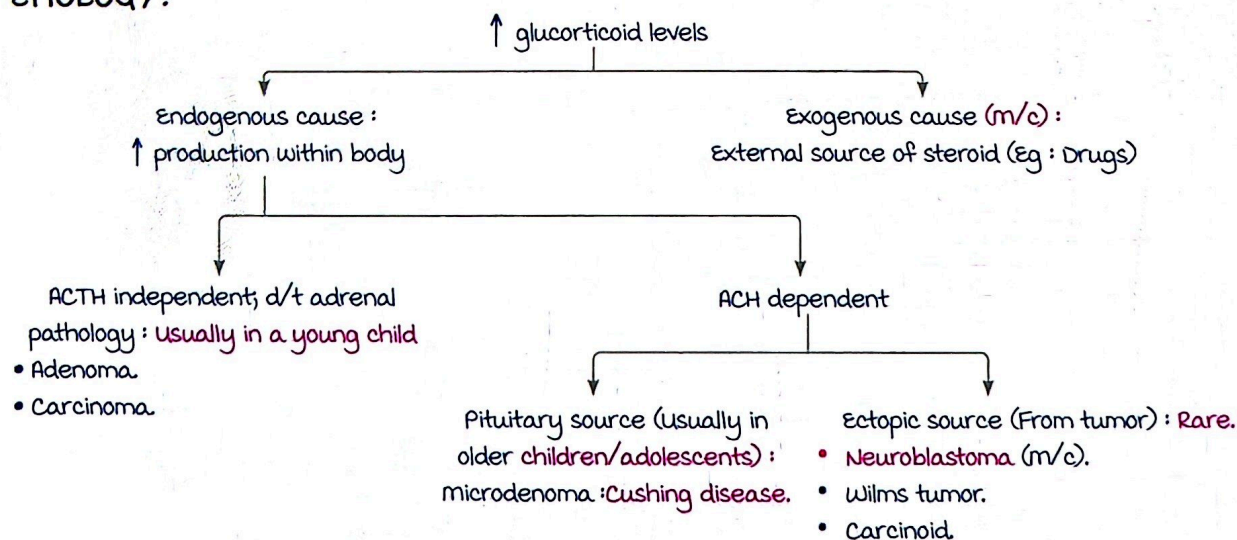
ACTH : Adrenocorticotrophic hormone.

Cushing's Syndrome

00:03:52

↑ glucocorticoids.

ETIOLOGY:



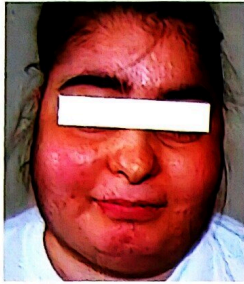
CLINICAL FEATURES:

↑ fat deposition : ↑ glucocorticoids → ~~×~~ Lipolysis → Fat deposition.

Feedback

----- Active space -----

• manifestations :



1. moon facies



2. Abdominal obesity



3. Striae :

4. Buffalo hump

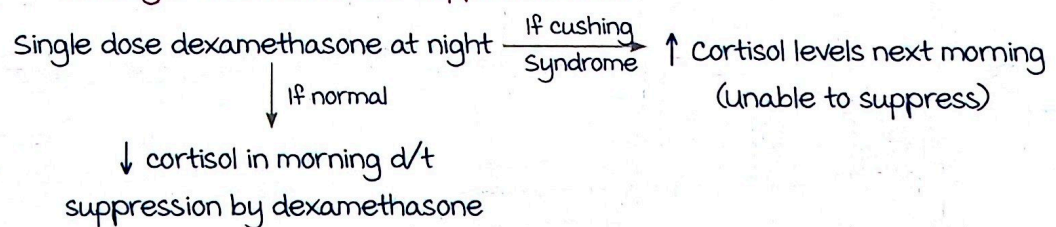
d/t excessive stretching of skin

- Hypertension : Intrinsic mineralocorticoid activity of some glucocorticoid.
- Hirsutism : Overstimulation of adrenal gland $\rightarrow$ $\uparrow$ production of androgens.
(In ACTH dependent conditions)
- Delayed puberty : Cortisol $\xrightarrow{\text{Inhibits}}$ Gonadotropin production.
- Proximal muscle weakness (Shoulders, hip).
- Short stature : Slowing of growth.

INVESTIGATIONS :

Screening :

- 24h- urine free cortisol levels.
- Overnight dexamethasone suppression test :



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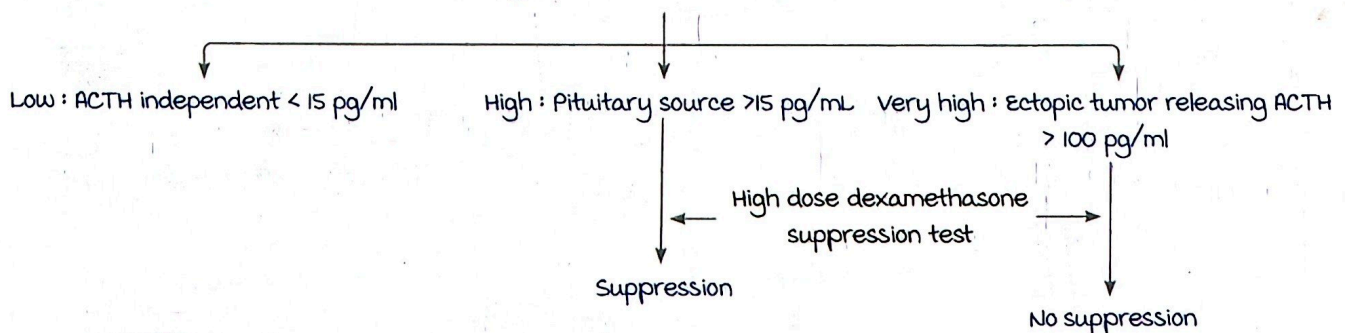
Diagnosis :

- Low dose dexamethasone suppression test :

5 $\mu\text{g/kg}$ of dexamethasone q6h $\times$ 2 days $\xrightarrow{\text{Cushing Syndrome}}$ $\uparrow$ cortisol level

$\downarrow$ F/B

ACTH estimation (in pg/ml) : To differentiate based on ACTH dependence.



TREATMENT :**Surgical :**

- Resection of identified tumor.
- Cushing's disease : **Trans-sphenoidal resection.**

medical :

- AKA medical adrenalectomy.
- medical inhibition of adrenal steroidogenesis by **anti-fungals** :
 - ketoconazole.
 - metyrapone.
 - metotane.

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Congenital Adrenal Hyperplasia (CAH)

00:18:50

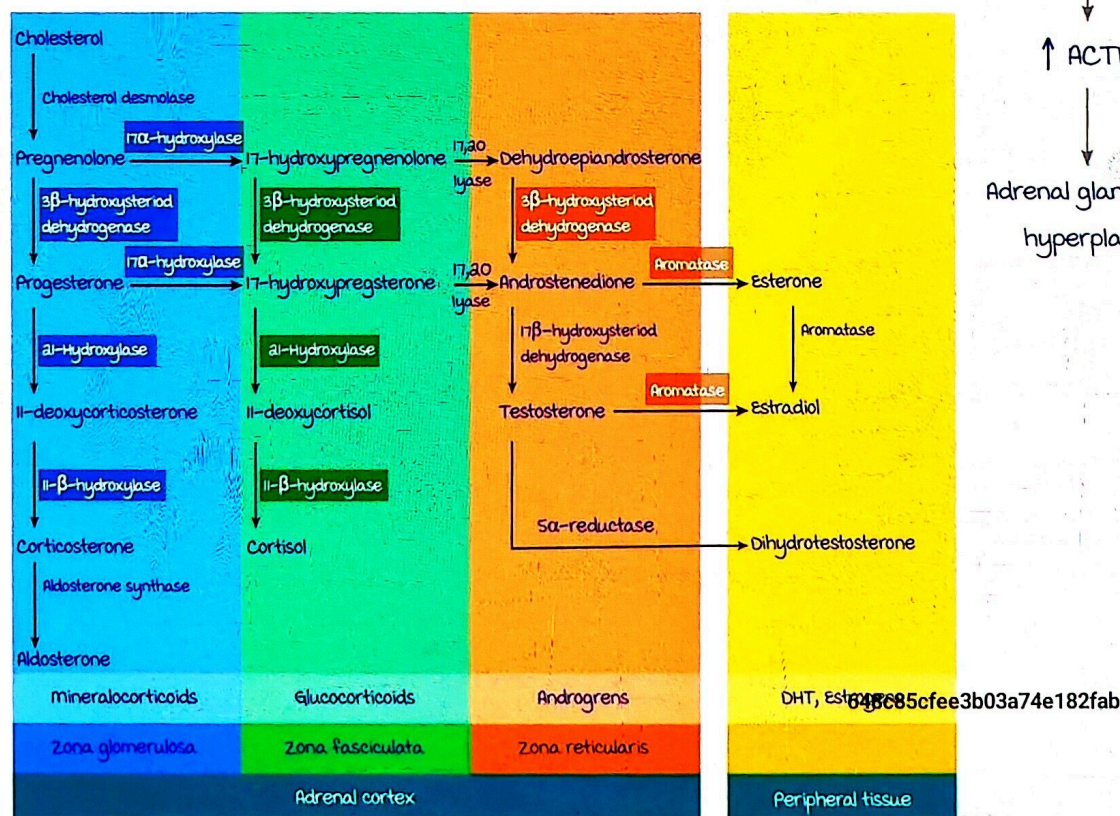
m/c adrenal disorder in childhood.

mode of inheritance : Autosomal recessive.

PATHOPHYSIOLOGY :

↓ enzyme → ↓ steroids (Glucocorticoid)

Feedback to pituitary gland

**Deficiency of enzymes :**

1. **21-hydroxylase (21-OH).**
2. **11 β -hydroxylase (11 β -OH).**
3. **17 α -hydroxylase (17 α -OH).**
4. **3 β -hydroxy steroid dehydrogenase (3 β -HSD).**

most common

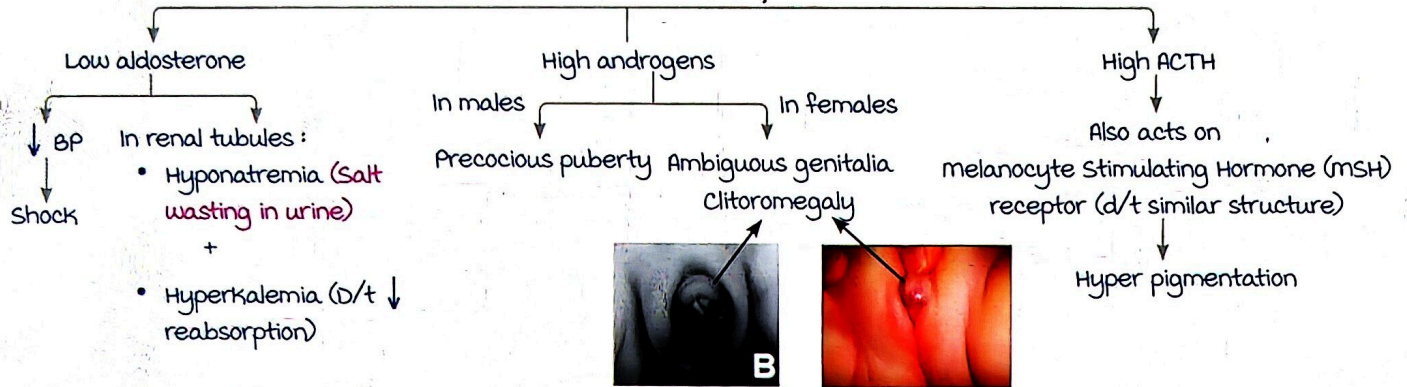
Least common

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21-HYDROXYLASE DEFICIENCY :

↓ enzyme → ↓ aldosterone + ↓ cortisol $\xrightarrow{\text{Feedback}}$ ↑ ACTH $\xrightarrow{++}$ ↑ androgens

khanirfan0392@gmail.com
D/t enzyme deficiency

Clinical features :**management :**

- Newborn screening : Estimation of 17-hydroxy progesterone (17-OHP) levels.
- Lifelong replacement of
 - Aldosterone : Fludocortisone.
 - Cortisol : Hydrocortisone.

Dose adjustment during stress (D/t ↑ requirement) :

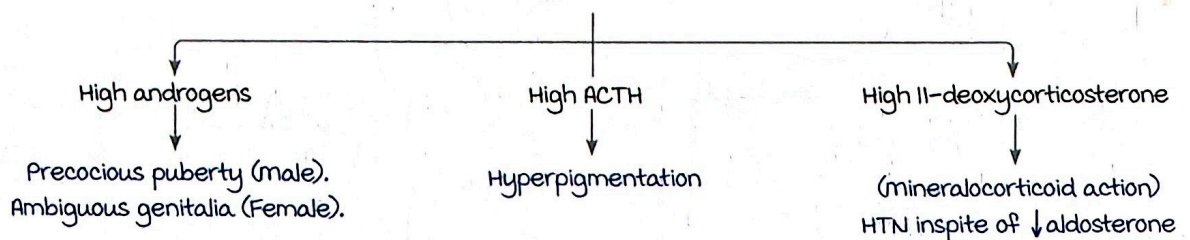
- minor stress : 2x to 3x dose (Infections)
- major stress : 4x to 5x dose (Surgeries)

- Surgical correction of ambiguous genitalia : Should be done within 2-6 months (Clitoroplasty).

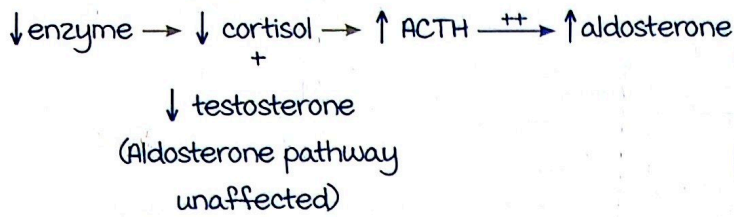
11-β-HYDROXYLASE DEFICIENCY :

↓ enzyme → ↓ Aldosterone + ↓ cortisol $\xrightarrow{\text{Feedback}}$ ↑ ACTH $\xrightarrow{++}$ ↑ androgens

d/t enzyme deficiency

Clinical features :

17 α HYDROXYLASE DEFICIENCY:



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Clinical features:

High aldosterone: Hypertension.

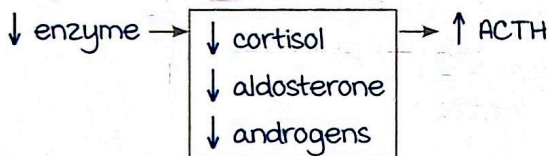
Low androgens:

- In males: undervirilization.
 - Hypospadias, bifid scrotum, delayed puberty.
 - In severe cases: Ambiguous genitalia.
- In females: usually asymptomatic.
 - Slow/delayed progression of puberty: d/t decreased aromatization of testosterone

High ACTH: Hyperpigmentation.



3- β -HSD DEFICIENCY:



Clinical features:

Low aldosterone: $\downarrow$ BP + hyponatremia + hyperkalemia.

High ACTH: Hyperpigmentation.

Low androgens:

- In males: undervirilization.
- In females: mild virilization d/t $\uparrow$ dehydroepiandrosterone (mild androgen) in older child.
- Features: Acne, hirsutism, menstrual irregularities.

POINTERS FOR DIAGNOSIS:

648c85cfee3b03a74e182fab		Serum androgen	
		$\uparrow$	$\downarrow$
Salt wasting features	+	21 OH deficiency	3 β HSD deficiency
	-	11 β OH deficiency	17 α OH deficiency

Feedback